

CS 2050 — Notes 09/16

Vacuous Proof

If we have $p \Rightarrow q$ and can prove that p is false, then $p \Rightarrow q$ is automatically true. This is a vacuous proof.

Trivial Proof

Whenever q is true, $p \Rightarrow q$ is true.

Proof of Equivalence

To prove that a theorem is a biconditional, prove that both $p \Rightarrow q$ and $q \Rightarrow p$ are true.